

Erdős Problem #856

EP-856

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and $f_k(N)$ be the maximum value of $\sum_{n \in A} \frac{1}{n}$, where A ranges over all subsets of $\{1, \dots, N\}$ which contain no subset of size k with the same pairwise least common multiple.

Estimate $f_k(N)$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>